Annex

The parameters shown in Table 3 represent key configurations that can be adjusted in the EQT tool for each execution method. These parameters allow users to customize the behavior of the tool based on specific requirements and computational resources. Below is a brief description of each parameter:

- **Estimate uncertainty:** Indicates whether the model should estimate the uncertainty associated with its predictions. This is particularly useful for assessing the confidence of the detections. It is only available for the complex method.
- **Number of Monte Carlo sampling:** Specifies the number of Monte Carlo simulations to perform for uncertainty estimation. A higher value provides more robust uncertainty estimates but increases computational cost. It is only available for the complex method.
- **Detection threshold:** Defines the minimum confidence score required for a detection to be considered valid. This parameter directly influences the sensitivity of the tool.
- **P threshold:** Sets the confidence threshold for identifying P-wave arrivals. A higher threshold reduces false positives but may miss weaker signals.
- **S threshold:** Sets the confidence threshold for identifying S-wave arrivals. Similar to the P threshold, this parameter balances sensitivity and specificity.
- **Use multiprocessing:** Enables or disables the use of multiple CPU cores for parallel processing, which can significantly speed up execution on systems with multiple cores. It is only available for the complex method.
- **Gpuid:** Specifies the GPU device to be used for computation. This is relevant in systems with multiple GPUs.
- **Gpu limit:** Defines the maximum GPU memory allocation for the process. Setting this value can help prevent memory overflows on systems with limited GPU resources.